Dynamic AI Exposure (DAX): pre-registered design memo v1

Status: v2 draft — PRIMARY DESIGN AMENDED 2026-08-18; not pre-registered

Draft date: 2026-08-06; amended 2026-08-18 (D1, D3, D4, F2)

Gate: `DAX-GATE1-memo`

Outcome-data status: SEALED until the signed tag `v1.0-preregistered`

This memo translates the DAX proposal and Amendment v1.1 into an executable design. The PI approved all 17 bracketed `[PI-DECISION]` defaults and all six non-numeric confirmations on 2026-08-06, and counter-signed four amendments on 2026-08-18 (section 11.2). They are frozen design choices, but the tag remains prohibited until the evidence checklist, power task, independent red-team, and rendered review are complete.

What changed on 2026-08-18, in one paragraph. The primary specification is no longer a stacked event study. Executing the previous section 3.2 against this memo's own registry left two estimable events where the power engine requires three, and the rule was non-monotone in evidence quality. The primary is now a continuous cumulative-dose design on the monthly index of section 2, with no event selection; the stack survives as secondary corroboration. The power pass bar, previously derived from the sample it judged, is now a frozen external constant. The primary estimand is stated explicitly as an incumbent margin. The previously registered entrant-margin companion is now exploratory after its pre-event feasibility audit failed. Five registry rows were demoted to `pending_second_date_locator` under a rule now enforced by the validator rather than applied by hand.

Review status. The independent cross-vendor red team of 2026-08-06 reviewed the superseded discrete design. Its `CONDITIONAL_GO` does **not** transfer to this draft, and the corresponding evidence-checklist item has been returned to unchecked. A fresh adversarial pass over this version is required before Gate 1.

0. Binding scope and feasibility conditions

The estimand concerns the U.S. occupation-level *full-task displacement frontier*: the wage-bill share of tasks for which AI substitution becomes privately cost-effective. It is not a net general-equilibrium employment effect and does not treat technical feasibility as realized adoption.

The signed 2026-07-10 feasibility decision binds this memo:

1. W4 capture of accessible historical model snapshots must finish before the

applicable 2026-10-23 and 2026-12-11 shutdown dates. 2. Retired vintages enter only through cited stand-ins, stand-in uncertainty enters the EIV analysis, and `gpt-4.5-preview` is excluded because no qualified stand-in was filed. 3. GDPval is referenced by task ID for internal research. No GDPval task text or derived task content enters W10a unless redistribution rights are clarified.

Primary task bundles, wages, and weights use 2021 O*NET/OEWS vintages. A 2019 wage baseline and annually updated *ONET bundles are robustness/decomposition variants, not replacements for the frozen primary index.

1. Event registry

1.1 Inclusion rule

An event enters the registry only if it changes at least one of: accessible model capability, input/output/reasoning-token price, or the set of model snapshots available for measurement. Every date and price requires two locators. Announcement-only events without API availability are recorded but do not receive an exposure dose until the API-effective date.

1.2 Machine-readable registry and source-complete core events

The frozen registry source is `dax/memo/event_registry_v1.csv`. It records analysis status separately from evidence status: a `candidate` is chronology, not permission to estimate an event. `python dax/memo/validate_event_registry.py` fails if IDs or dates are malformed, either locator is absent, sources repeat, or an event marked `eligible` lacks verified status. The validator does not pretend to read source content; rows marked `pending_second_date_locator` remain ineligible until a second source independently dates API availability.

`event_table_shell_v1.csv` is the downstream continuous-dose reporting shell. Its continuous-dose distribution, residualized-variance, and leave-one-event- out fields are intentionally blank with an explicit W5 fill status until a real W5 panel exists. Blank values are deferrals, not zeroes or completed measurements, and the shell cannot be used as empirical evidence before W5.

Event ID	API-effective date	Classification	Evidence 1	Evidence 2	Status
`GPT4_LAUNCH`	2023-03-14	capability	OpenAI, GPT-4	OpenAI, GPT-4 technical report	date verified; price-history row pending W2
`GPT4O_LAUNCH`	2024-05-13	capability + price	OpenAI, Hello GPT-4o	OpenAI, GPT-4o product release	lead price event; official release states API cost was 50% lower than GPT-4 Turbo
`O1_PREVIEW`	2024-09-12	capability	OpenAI, Introducing o1-preview	OpenAI API changelog (September 2024)	date verified
`GPT41_LAUNCH`	2025-04-14	capability + price	OpenAI, Introducing GPT-4.1 in the API	OpenAI API changelog (April 2025)	date and launch prices verified

The current registry contains 21 dated chronology rows through 2026-08-06: four currently eligible core events, sixteen candidates, and the binding `gpt-4.5-preview` exclusion. Four candidates are explicitly pending a second independent date locator. Candidate status is resolved before Gate 1; W5 later applies the signed dose threshold without adding or deleting chronology rows. The mechanically fillable event-by-event table is frozen in event_table_shell_v1.csv. W5 may fill only its blank window, crossing-count, dose-distribution, and effective-weight fields after applying Decisions 1--3; it may not change event IDs, dates, thresholds, or column definitions. At the Gate-1 freeze, any candidate still lacking two independent dated locators or a source-complete dated price row is mechanically reclassified source_failed and excluded from the primary event set. It remains in the chronology with the failed source fields and exclusion reason; it cannot be restored after outcomes open except through a dated deviation memo. This rule includes every unresolved conflict_b: W2 must select the price row supported by two independent dated locators and archive the conflicting rows, or the event becomes source_failed; averaging or choosing the lower price is prohibited. If W4 cannot recover either a qualifying historical snapshot or a feasibility-approved cited stand-in, the event is likewise excluded as measurement_failed and retained only in the descriptive chronology. Registry price status relative_price_verified means that two locators verify a relative price change but W2 has not yet attached the absolute pre/post price rows needed for dose construction; it is therefore not source-complete at Gate 1 unless W2 supplies those rows. pending_w2 has no verified price change, and n_a is allowed only for a capability-only event whose price is unchanged.

1.3 Registry completion queue

Candidate rows now cover GPT-4 Turbo/DevDay, GPT-4o mini, full o1, o3-mini, o3/o4-mini, GPT-5-family launches through GPT-5.6, and located price changes. They remain non-filed analysis events until the CSV records two date locators and W2 attaches dated prices. W2 emits the source-complete price-history panel; W1 imports its event IDs without changing the rules below.

The current official deprecations page contains both a historical 2025-09-26 row saying gpt-4-1106-preview shut down on 2026-03-26 and a newer 2026-04-22 row listing an October 23, 2026 shutdown for the same slug. The signed feasibility adjudication uses the newer row for capture planning. The memo preserves both locators as a provenance conflict; W4 must test actual availability before budgeting the row.

[PI-DECISION 1] *(re-scoped under D1, 2026-08-18)* **Event eligibility.** Originally this threshold governed inclusion in the primary stacked analysis. The primary specification no longer selects events, so it now governs *separate reporting*: an event with at least one percentage point of occupation wage-bill dose at the median occupation is named in the descriptive chronology and receives its own leave-one-event-out diagnostic. Smaller events still enter the DAX dose path in full. **No event is dropped from the dose path on account of its size**, which is the substantive change — the old rule discarded small events from estimation, and the continuous design does not.

2. DAX construction objects fixed for the design

For task t, occupation o, month m, mapping k, cost variant v, deployment parameter delta, and failure-cost rule g, define

```
pi_eff = 1 - delta * (1 - pi)
```

and crossing indicator

```
A_tom = 1[c/pi_eff + f*(1-pi_eff)/pi_eff < w].
```

DAX_om is the frozen-2021 wage-bill share of occupation tasks with A_tom=1. DeltaDAX_oe is the share newly changing from zero to one at event e. A task cannot cross twice in the primary monotone chronology. If a measured capability regression or price increase reverses feasibility, the reversal is retained as a signed change in a separate non-monotone diagnostic and does not erase its first crossing.

Primary grid:

- mapping: GDPval, Tolan-style, Eloundou-style;
- cost: API list, 0.6 * list, open-weight marginal;
- delta: 1.0, 0.8, 0.6;
- failure multiple by O*NET consequence tier: 0.25, 1, 4 times task wage;
- failure sensitivity: zero, half, primary, and double tier multiples;
- cost accounting: billed tokens and completion-only effective tokens for

reasoning models;

- wage vintage: 2021 primary, 2019 alternative.

The flip rate is the share of occupation-event cells whose crossing status changes anywhere across the failure-cost grid. The already-proposed trigger is strictly greater than 20%. Crossing this trigger halts W5 before outcome work and returns the threshold design to the PI; it does not authorize choosing the best-performing grid point.

3. Continuous cumulative-dose protocol

Design change, D1, PI-approved 2026-08-18. This section previously specified a stacked common-event design with clean windows. Executing that specification against this memo's own registry left two estimable events from the four currently eligible rows, and one if the registry is completed as planned; the power engine refuses to run below three. The rule was also non-monotone in evidence quality, so completing the registry made identification worse and the project's own W2 sourcing work self-defeating.

The cause was not an implementation error. A stacked event study with clean windows presumes discrete, well-separated shocks; with 21 events across 41 months the treatment is effectively continuous, and every discrete repair either collapsed to a single event or manufactured purity by ignoring events that actually occurred. The full option comparison is reproducible via python dax/memo/validate_window_survival.py --options and recorded in dax/memo/PI_DECISION_D1_2026-08-18.md.

3.1 Unit, dose, and comparison

The analysis unit is person-month in IPUMS-CPS. The treatment is the DAX level DAX_om of the person's reference occupation in the reference month, assigned through the employment-weighted many-to-many CPS to O*NET-SOC crosswalk.

There is no event selection, no stacking, and no clean-window rule. Every source-verified event contributes to the dose path

```
DAX_ot = DAX_o,0 + sum over events e with API-effective month <= t of DeltaDAX_oe
```

and identification comes from occupations accumulating dose on different profiles across the same calendar months. Comparison is a continuum rather than a control arm: conditional on the controls in section 9.1, the contrast is between occupations whose dose rises faster and those whose rises more slowly. Occupations with zero cumulative dose sit at the low end of that continuum.

Consequence for W2, stated because it reverses the previous incentive. Under the superseded rule, each newly verified event shortened clean windows and reduced the estimable set. Under this design each newly verified event adds dose variation and can only increase identifying variation. Registry and price work now improves the design monotonically.

3.2 Parameters of the dose path

The Decision numbers below are retained rather than renumbered so the PI approval record of 2026-08-06 stays aligned. Where D1 changed a decision's role, that is stated explicitly rather than silently reinterpreted.

Decision 1 (section 1.3) is re-scoped rather than repeated here: it now selects which events are *separately narrated*, not which enter the dose path. Every source-verified event contributes dose regardless of size.

[PI-DECISION 2] *(retired from the primary under D1)* **Window and stacking parameters.** The primary specification has no windows, so this decision no longer binds it. The rule survives unchanged as the governing parameter of the secondary stacked corroboration in section 9.4, at the four-month compound setting. The number is **not** reused for anything else.

[PI-DECISION 3] Minimum occupation dose. Unchanged in substance and now applies only to descriptive treated-bin summaries: a treated occupation-event cell requires `DeltaDAX >= 0.01`. The continuous regressor retains all positive doses, including those in `(0, 0.01)`.

[PI-DECISION 4] *(reversed under D1)* **Dose accumulation.** The superseded design used the event increment `DeltaDAX_oe` as the primary regressor and the level as a control, in order to isolate new information at a discrete event. With no discrete events the reasoning inverts: the primary regressor is the cumulative **level** `DAX_ot`, which is the object the index in section 2 actually constructs, and the within-month increment is reported alongside it as a secondary specification. This reversal is the substantive econometric content of D1 and is flagged here so no reader infers it from the equations alone.

[PI-DECISION 5] *(retired under D1)* **Earlier crossers.** This decision existed to say whether previously-treated occupations could serve as controls inside a later clean window. There are no clean windows and no control arm, so the question does not arise. What survives from it is the monotonicity convention: a task that has crossed stays crossed in the primary path, and a measured capability regression or price increase is recorded as a signed change in the separate non-monotone diagnostic rather than erasing the first crossing. Occupations with high pre-period dose are not excluded; pre-event DAX level enters as a registered control and a robustness sample excludes occupations already above the median dose at 2023-02.

3.3 What the secondary stack retains

The compound-event state algorithm of the superseded draft is retained verbatim for the secondary specification: order components by exact API-effective timestamp then event id, define the pre state immediately before the first component and the post state immediately after the last, recompute crossings on those two states, and set the composite `DeltaDAX` to the wage-bill share moving from zero to one. It is a correct piece of machinery that the primary design simply no longer needs.

4. Identifying conditions and estimands

IC-1 — Conditional parallel trends in a continuous, time-varying dose. For every pair of calendar months and every pair of dose paths in the common support, the expected change in untreated potential outcomes is equal across paths, conditional on occupation effects, calendar-month effects, industry-by-month effects, frozen static-exposure decile interacted with month, occupation interest-rate sensitivity, and the registered baseline covariates. This is the continuous-treatment condition of Callaway, Goodman-Bacon and Sant'Anna (2024); it is strictly stronger than the two-period version, because it must hold at every month rather than only across a single window.

IC-2 — No dose-correlated interference. Conditional on the same controls, spillovers from high-dose to low-dose occupations do not vary with dose. Mobility-network spillover estimates are auxiliary and do not repair a violation.

IC-3 — Mapping timing exclusion for the secondary IV. Conditional on controls, errors in the Tolan and Eloundou doses are independent of the GDPval mapping error and of untreated outcome innovations. Shared structural errors, such as all three mappings equating benchmark performance with production success, violate this condition and are not removed by instrumenting.

IC-4 — Frozen entry mix (exploratory entrant analysis only). The entry-occupation distribution `pi_go`, estimated once over 2021-11 to 2023-02, is independent of post-2023 innovations in entrant employment conditional on cell and month effects. If entrants systematically divert away from exposed occupations after the pre-period, that diversion is an outcome and must not enter the regressor; freezing `pi_go` in the pre-period is what enforces this, and the assumption that the frozen mix remains a valid *instrument* for exposure, rather than a description of realised entry, is the companion's central identifying requirement. The pre-event audit failed the registered occupation-level estimability condition, so this assumption is not used for confirmatory inference and the companion remains outside Gate 1.

Estimands

Primary (incumbent margin). The change in employment probability and in unconditional weekly hours associated with a 0.10 increase in occupation DAX level, among young workers with a prior occupational attachment, over the observed common-support population. Reported pooled, by 12-month calendar block, and as a binned nonparametric dose response.

Exploratory entrant margin. A descriptive analysis may relate employment among clean, adjacent-month linked nonemployment-to-employment transitions to coarse pre-event entry-mix exposure. It is not a registered causal estimand and cannot support a headline, confirmatory, or Gate-1 claim.

The primary incumbent estimand does not cover labour-market entry. The clean linked-transition subset used in the exploratory analysis is not the exact complement of the primary sample and must not be described as partitioning the 22--25 population.

Why not a single pooled estimand. Merging the two would require imputing an occupation for people who never held one. That is fabrication, and it would produce a coefficient that means neither margin. It is rejected under every budget.

5. Mapping hierarchy and measurement quality

GDPval is primary. Tolan and Eloundou mappings are independent robustness constructions. All task matches retain similarity, coverage, confidence, adjudication status, and a top-quartile match-quality flag. Unmatched tasks remain explicit and their wage-bill share is reported.

The pre-existing survive-all-three rule is binding: the point-estimate sign is consistent across all three mappings and the relevant test rejects at the 10% level under at least two mappings.

[PI-DECISION 6] Mapping tiebreaker default. If all signs agree but sampling precision differs, report GDPval as primary and the median standardized point estimate across mappings as the descriptive synthesis. The tiebreaker never overrides a sign conflict or the survive-all-three failure rule.

[PI-DECISION 7] Human-validation targets. Audit 10% of mapping-C annotations stratified by occupation family, score decile, and ambiguity flag. Require weighted Cohen's kappa of at least 0.70 and at least 90% agreement on the binary crossing-relevant label before W5. Failure returns the rubric for redesign before index construction.

6. Index validation and behavioral first stage

6.1 Public index validation

[PI-DECISION 8] Convergent-validity threshold. The primary GDPval/list/ `delta=1` DAX level must have occupation-level Spearman rank correlation of at least 0.50 with the frozen Felten-Eloundou-Webb ensemble at one or more pre-registered benchmark months, with Kendall correlation reported. This is a moderate convergence requirement: lower suggests construct failure, while a near-one requirement would eliminate the intended dynamic/cost distinction.

Event alignment, capability-versus-price decomposition, crossing-order stability, and comparison with published usage indices are reported without post-hoc pass thresholds except those explicitly added by PI signature.

6.2 Usage-share first stage

The named human team alone runs frozen code on real OpenAI aggregates. The agent-developed pipeline sees synthetic data only.

The first-stage unit is frozen O*NET task-or-IWA cell by event by calendar month. For cell `i` and month `t`, within-month usage share is the number of classified eligible ChatGPT messages assigned to `i` divided by all classified eligible messages in the same negotiated product population and calendar month, before cell suppression. For event `e`, the pre-event mean is the simple arithmetic mean of that share over every surviving clean pre month for which the cell is observed; the complete-case rule in Section 8 requires observation in all required pre and post months. The normalized outcome is `100 * (share_iet / premean_ie - 1)`. If `premean_ie=0`, the normalized outcome is undefined and the cell is excluded from the normalized first stage rather than regularized. Its raw share path is reported as an auxiliary diagnostic, along with the excluded task-mass share.

[PI-DECISION 9] Minimum behavioral effect. Success requires the pooled post-crossing coefficient on that normalized outcome to be at least 5, meaning a 5% relative usage-share increase, with its 90% confidence interval excluding zero. Event-time coefficients remain required diagnostics. Asking-to-Doing composition is a mechanism outcome; default materiality is a two percentage-point increase in Doing share, reported with uncertainty but not a separate gate.

A positive first stage supports dateable translation from feasibility to behavior. A null does not disprove technical feasibility; it localizes adoption away from measured ChatGPT message channels or indicates that unit economics is not the binding adoption margin.

7. CPS sample, outcomes, and power specification

7.1 Population and the estimand this sample supports

Population: U.S. persons ages 22-25 in monthly IPUMS-CPS, November 2021 through the latest frozen month, using person weights. Dose is assigned at the CPS occupation code after employment-weighted crosswalking to O*NET-SOC.

A person's reference occupation is the most recent valid CPS occupation observed for the same `CPSIDP` strictly before the reference month and no more than 15 calendar months earlier. That reference is held fixed for the person's contribution in that month. This preserves respondents who are unemployed at the time of measurement and prevents a post-treatment occupation change from redefining treatment. A registered robustness check caps the lookback at nine months.

[D4 Part 1, PI-approved 2026-08-18] The estimand is stated, not implied. The lookback rule excludes every person with no prior occupational attachment. For ages 22-25 that is, specifically, labour-market entrants. Therefore:

> The primary estimand is the effect of AI exposure on the employment and > hours of young workers **who already hold an occupational attachment**. It > is an incumbent margin. It is not an effect on labour-market entry. Persons > with no pre-period occupation observation are outside the estimand by > construction, not missing data.

The count, weighted share, and lookback-duration distribution of excluded persons are reported in the main text, not an appendix.

The proposal's 13% benchmark is drawn from literature in which the effect operates substantially through reduced **hiring**. That benchmark and this primary estimand are therefore not the same object. A null on the incumbent margin is not evidence against an entrant-margin finding, and the memo does not present it as such.

7.2 Entrant-margin analysis (demoted to exploratory 2026-08-18)

The PI approved a registered secondary companion on 2026-08-18. The required pre-event audit then showed that the design is not estimable at the registered occupation-by-cell level. The controlling red-team adjudication therefore demoted it to exploratory; this is an evidence-triggered deviation from D4 Part 2, not a new affirmative PI choice.

The benchmark literature's reduced-hiring mechanism still makes entry an important descriptive margin, but motivation does not overcome failed measurement and support gates. The registered complement rule is invalid because it mixes true entrants, CPSIDP linkage failures, rotation entry, and long nonemployment.

The private audit instead examined adjacent-month CPSIDP-linked nonemployment-to-employment transitions in eligible rotation months. It found 1,623 linked entries across 16 demographic-education cells and 822 cell-occupation pairs; the pair-count median was 1, the maximum 18, and 100% of entries were in pairs below the diagnostic minimum of 20. Expected-prior-interview link failure was 16.3% unweighted (16.5% weighted). Therefore occupation-level `pi_go` is not estimable as registered.

An exploratory analysis may construct a coarse pre-event entry-mix-weighted DAX,

`EDAX_gt = sum_o pi_go * DAX_ot`,

where `pi_go` is estimated only from the pre-event window and never updated with post-treatment entry choices. Any pooling, shrinkage, cell definition, and sampling-error propagation is exploratory unless separately approved prospectively. Results must be labeled descriptive/exploratory, show the sparse cell distribution, and may not be used to repair, confirm, reject, or headline the primary incumbent result. No entrant power table enters Gate 1. Restoring a registered companion requires a PI-approved entrant definition, pooling threshold, and sampling-error propagation rule followed by fresh red-team review.

7.3 Outcomes

Primary: (1) employment indicator; (2) unconditional usual weekly hours, coded zero for the non-employed. Secondary: log hourly wage among the employed, and conditional weekly hours. Auxiliary: occupation switching, unemployment reason, search duration. The college-attainment split is secondary, receives its own power table, and cannot support headline claims.

[PI-DECISION 10] Multiplicity default. Unchanged: Holm across the two primary outcomes at two-sided 5%, unadjusted intervals reported alongside adjusted p-values, Benjamini-Hochberg at 10% applied separately to the secondary and auxiliary families. Exploratory entrant analysis is outside the registered outcome families and is labeled non-confirmatory; it cannot borrow the primary family's error budget or enter headline multiplicity claims.

7.4 Power standard

[PI-DECISION 11] *(numeric part superseded by D3, PI-approved 2026-08-18)* The original bar was `min(6.5pp, 0.5 x baseline_employment_gap)`. The gap term was estimated from the analysis sample, so it moved with the event set: dropping a single event loosened the bar by 185% while the estimator got 19% worse. A standard that moves with the specification it judges cannot fail.

The standard will be a frozen absolute constant once its real pre-event CPS baseline is frozen:

`ceiling = max_mde_fraction x relative_decline x baseline_level`

where the prospectively PI-selected primary `relative_decline` is **0.13**. This is an external empirical calibration scale from the authenticated 2025-08-26 authored version, not an estimate of the DAX treatment coefficient. It is a relative decline in EMPLOYMENT (headcount), not in payroll. "Payroll" names the data source, not the outcome: the estimate comes from ADP administrative payroll records. The project's own proposal states it unambiguously — "U.S. payroll data, by contrast, show a 13 percent relative employment decline among workers ages 22-25 in the most AI-exposed occupations" (`docs/DAX_ERE_Proposal_v3.md:12`, citing Brynjolfsson, Chandar and Chen 2025, "Canaries in the Coal Mine? Six Facts about the Recent Employment Effects of Artificial Intelligence", `docs/DAX_ERE_Proposal_v3.md:100`). Because it is a *relative* decline, the absolute percentage-point benchmark is `0.13 x baseline employment rate`.

Prespecified interpretation and sensitivities. PI decision commit `4577fecab7b4e142cb28d78d4aec0800637c7b05`, made before real power results, sets 0.16 from the authenticated 2025-11-13 revision as a version-update empirical sensitivity. The historical 0.19 value is retained only as a normative PI design-target sensitivity; its external empirical provenance remains unresolved and it is not a literature estimate. The external paper and DAX differ in exposure definition, sample, period, treatment unit, and estimand, so none of these values is represented as a directly comparable DAX causal coefficient. The value may not be changed after power results are seen.

With `max_mde_fraction = 0.5`, and `baseline_level` the pooled person-weighted pre-event level of the outcome (employment rate; mean unconditional hours) computed **once** over 2021-11 to 2023-02 and never recomputed. The window ends the month before the first eligible event, so the constant cannot contain post-treatment information, and it requires neither a dose definition nor an event set.

The constant lives in `dax/memo/power_calcs/power_standard.json` with its provenance and a `status` field. Until it is `FROZEN`, both power engines report minimum detectable effects and return `adequately_powered: null`. They do not guess and they do not pass.

Recomputation after the event set, mapping, dose definition, or sample changes is forbidden. Failure does not authorise sample or specification search: it triggers the proposal's informative-power limitation and PI reconsideration, exactly as the original Decision 11 provided.

Unit of estimation versus unit of simulation (M5, resolved 2026-08-18). The *estimator* is person-month: section 9.1 is estimated on person records, and the registered person covariates `X_i` enter there. The *power simulation* operates on occupation-month cell moments, because cell moments are what the frozen pre-event fixture provides and because the variance of `beta` is driven by the occupation cluster structure rather than by within-cell person variation. These are not the same object and the memo does not pretend they are.

No ordering between the cell and person-level MDE is asserted. Omitting `X_i` can raise residual variance, but aggregation also removes within-cell outcome variation and changes weights and leverage; without additional assumptions those forces do not sign the difference. The cell result is therefore only a smoke-test/planning calculation and can neither pass nor fail Gate 1.

The Gate-1 calculation runs the person-month estimator directly on the frozen pre-event extract, with post-event outcomes simulated against W5's real dose panel. Its occupation-clustered rejection rates and MDEs are the only power result used for the gate.

The power simulation must model the continuous design on pre-event moments only, preserving occupation cluster sizes, person weights, crosswalk dose dispersion, the dose path, and the observed employment-hours covariance. The engine refuses to run if the moment file contains any month at or after the first event.

[PI-DECISION 12] Dispersion flag. Unchanged: flag a CPS code as low quality when the weighted within-code standard deviation of O*NET dose exceeds 0.10 or the maximum mapping weight is below 0.50, and re-estimate on the complement as a registered robustness check.

8. Suppression, privacy, and minimum estimability

Suppressed OpenAI aggregate cells are missing by design and are never filled with zero, midpoint, model prediction, or neighboring-cell values. Primary first-stage estimation uses only cells observed in every required event-time period. An inverse-observation-probability weighted sensitivity analysis may use only publicly described suppression predictors and is labeled secondary.

[PI-DECISION 13] *(scope clarified under D1)* **Minimum estimability.** This condition governs the **OpenAI usage first stage only**. It was written for the first stage and the pre-D1 power engine wrongly enforced clause (c), the three-event minimum, as a global gate on the CPS power calculation; under the continuous primary there is no event count to gate on, and the continuous engine does not apply it. Clause (c) continues to bind the first stage, where events are genuinely the unit of estimation. Interpret the first stage only if: (a) observed cells cover at least 70% of the wage-bill-weighted crossed task mass; (b) at least 50 occupation/task cells remain; (c) at least three eligible events remain; and (d) no single event contributes more than 50% of the effective estimation weight. For event `e`, that weight share is `sum_i w_ie * x_tilde_ie^2 / sum_j sum_i w_ij * x_tilde_ij^2`, where `w` is the event-normalized CPS person weight and `x_tilde` is the primary continuous dose-by-post regressor residualized on the complete frozen nuisance design. Otherwise report selection and MDEs without an effect interpretation.

Only the PI and named RA may open or run code on real usage aggregates. The repository stores schemas, synthetic data, frozen code, and aggregate cleared diagnostics only—never cell-level NDA data or derivatives.

9. Econometric implementation and diagnostics

9.1 Primary estimating equation

For person `i` in occupation `o`, industry `j`, and calendar month `t`:

```
y_iojt = beta * (DAX_ot / 0.10) + alpha_o + tau_t + gamma_jt + delta_{d(o),t} + X_i'theta + e_iojt
```

where `alpha_o` are occupation effects, `tau_t` calendar-month effects, `gamma_jt` industry-by-month effects, `delta_{d(o),t}` frozen static-exposure-decile-by-month effects, and `X_i` the registered person covariates. Standard errors cluster on the original CPS occupation code. `beta` is the coefficient per 0.10 DAX increment.

Occupation effects absorb level differences and month effects absorb the common time path, so `beta` is identified by the interaction of occupation-specific dose magnitude and timing with the calendar. This is the sense in which the design is a continuous difference-in-differences rather than a cross-sectional exposure regression.

9.2 The degeneracy diagnostic

If every occupation's dose path were proportional to a single common path, `DAX_ot = theta_o * f(t)`, the design would collapse to one exposure-times-post contrast and the timing variation would contribute nothing. This is a measurable property of the dose matrix, not a matter of opinion, and it is reported before any estimate:

Required diagnostic — dose-profile reporting. Report the effective rank of the demeaned occupation-by-month dose matrix and the share of its variation in the leading singular component. A leading share above 0.95 with effective rank 1 is a degenerate design: report it as such, interpret `beta` as a single exposure contrast, and do not present event-time language. The threshold is a reporting trigger, not a pass/fail gate, and no estimate is withheld on the basis of it.

Consequence, pre-registered (M6, resolved 2026-08-18). Deciding what a degenerate design means *after* seeing the leading share would be a specification choice made with knowledge of the data, so the consequence is fixed now. If the design is degenerate:

1. The paper does not claim to identify a *dynamic* exposure effect. The

headline claim becomes a cross-sectional exposure contrast estimated with panel controls, and every use of "timing", "event", or "dynamic" is struck from the claims about `beta`. 2. The DAX index's contribution over existing static measures is then argued on the **crossing chronology** — which tasks cross when, and at what price — not on the regression, since a degenerate dose matrix means the regression is using little more than a static ranking. 3. The Decision 8 convergent-validity result becomes load-bearing rather than descriptive: if DAX is behaving as a static measure, its rank correlation with the Felten-Eloundou-Webb ensemble is the evidence that it is measuring the intended construct at all. 4. The degeneracy statistic is reported in the abstract-level summary, not buried in diagnostics.

A design that is degenerate is still publishable. It is a different paper, and saying which paper in advance is the point.

This is a required diagnostic rather than a numbered PI decision: it triggers reporting language, never inclusion or exclusion, so it commits the PI to no threshold. The 17 approved decisions are unchanged in number and membership.

9.3 Required diagnostics

- **placebo-lead pre-trend test** over 2021-11 to 2023-02 (see Decision 14);
- event-time-style plot in calendar time: `beta` estimated separately by

12-month block, with the pre-event blocks shown;

- binned dose-response using bins frozen from the pre-event dose distribution;

- common-support and leverage plots by occupation and by decile;
- with and without industry-by-month effects;
- within-static-decile dose and industry-composition correlation;
- leave-one-occupation-out and leave-one-decile-out estimates;
- leave-one-event-out re-derivation of the dose path, which under the

continuous design perturbs the path rather than deleting a stack;

- Rambachan-Roth sensitivity at M-bar 0.5, 1, and 2 plus the

relative-magnitudes restriction, with M-bar 1 as headline;

- placebos: FOMC dates, permuted mappings, backward-shuffled price histories,

and a dose path shifted forward by 12 months.

[PI-DECISION 14] *(re-specified 2026-08-18 — the previous form was not computable)* **Pre-trend test.**

The superseded text required "a joint test of all pre-event dose coefficients". Under the continuous design the regressor is cumulative `DAX_ot`, which is **identically zero for every occupation** throughout 2021-11 to 2023-02: the first eligible event is 2023-03. A regressor with no variance has no coefficient, so that test could never have been run. It is replaced, not weakened.

The pre-trend test is a **placebo lead on eventual exposure**. Let `D_o = DAX_o,2024-12`, the occupation's cumulative dose at a frozen horizon, which is a fixed occupation characteristic and therefore *does* vary in the pre-period. Restricted to pre-event months only, estimate

```
y_ot = phi * D_o * (t - t_0) + alpha_o + tau_t + gamma_jt + e_ot
```

and test `phi = 0`. This asks the question that matters: were occupations destined for high exposure already on different trajectories before any exposure existed? A non-zero `phi` is the violation of IC-1, and unlike the superseded test it is estimable.

The design passes the conventional pre-trend diagnostic only if `phi` has p >= 0.10 and no single pre-period block interval excludes zero after Holm correction. The horizon 2024-12 is frozen here so that `D_o` cannot be chosen after seeing the result; the test is also reported at 2023-12 and 2025-12 as registered robustness, with the 2024-12 version as headline.

The pre-event window is approximately sixteen months, which is short. This diagnostic is therefore weaker here than in a long pre-period, HonestDiD sensitivity remains mandatory regardless of the result, and the shortness is reported as a limitation rather than absorbed silently.

9.4 Secondary stacked corroboration

The discrete stacked design of the pre-D1 draft is retained as corroboration under the compound rule at four months. It is reported with its own power statement, its estimable-event count, and an explicit statement that it is not confirmatory. Where primary and secondary disagree, both are reported and the disagreement is discussed; the specification that produces the larger or more significant estimate is not selected.

10. EIV, cross-mapping IV, and deployment calibration

10.1 Errors-in-variables simulation

For each mapping, perturb task success probability using distributions calibrated to human disagreement and item-level score dispersion, redraw at least at task and occupation-correlated levels, and rederive crossings.

[PI-DECISION 15] EIV acceptable bounds. Before outcome work, require median absolute crossing-date shift no greater than one month, 90th-percentile shift no greater than three months, no more than 10% of occupation-event cells changing dose bin, and implied linear attenuation no worse than 0.80. In each EIV draw, construct the exact frozen stack and residualize both the unperturbed dose-by-post regressor `x_true` and perturbed regressor `x_obs` on the same nuisance matrix with the same event-normalized weights. The draw-specific attenuation factor is the weighted slope `Cov_w(x_obs_tilde,x_true_tilde) / Var_w(x_obs_tilde)`: the coefficient obtained from outcome-on-observed-dose when the true coefficient is one and no sampling noise is added. The 0.80 bound applies to the median factor across draws; report its 5th--95th percentile interval. Exceeding any bound does not permit selecting a quieter mapping; it triggers redesign or an explicitly weakened estimand.

10.2 Cross-mapping IV

Instrument GDPval `DeltaDAX` separately and jointly with Tolan and Eloundou doses. Report reduced forms, first stages, overidentification diagnostics, and the IV/OLS ratio as a reliability diagnostic.

[PI-DECISION 16] Weak-instrument rule. Interpret IV magnitudes only when the heteroskedasticity/cluster-robust effective first-stage F statistic is at least 10. Otherwise report weak-instrument-robust intervals and treat IV as uninformative.

10.3 Post-first-stage delta calibration

A raw ratio of observed to predicted usage jumps is not, by itself, a valid estimate of `delta`, because `delta` changes `pi_eff`, crossing sets, and predicted event jumps nonlinearly. DAX v1.1 therefore uses the pre-registered minimum-distance estimator

```
delta_hat = argmin_{d in [0,1]} sum_e weight_e * (J_obs_e - J_pred_e(d))^2,
```

where `J_pred_e(d)` is recomputed through the full crossing pipeline and weights are inverse estimated variances of observed jumps. Bootstrap resamples the eligible usage cells and events. This calibration is post-first-stage and cannot change this paper's primary grid `{1.0, 0.8, 0.6}`. Estimate each jump variance with a pre-registered two-way pigeonhole bootstrap: 1,000 draws independently resample eligible task/IWA cells and eligible events with replacement, rerun the frozen first stage, and use the sample variance of the resulting event-specific jump. Set weight to the reciprocal of that variance, winsorized at the frozen 1st and 99th percentiles; a zero or non-estimable variance excludes that event from calibration and is reported.

[PI-DECISION 17] Calibration reporting. Search a 0.01 delta grid on `[0,1]`, report the minimizer and percentile 95% bootstrap interval, and label estimates boundary-limited when the minimizer is 0 or 1. Do not report delta_hat if the minimum-estimability condition fails.

11. Frozen output and deviation policy

W5 must emit the full mapping × cost × delta × failure-grid panel, crossing chronology, event table, flip-rate report, EIV diagnostics, capability-only counterfactual, distributional variant, and live-vintage decomposition. No configuration may be dropped because it performs poorly.

Any post-tag change to an event, window, dose, threshold, mapping hierarchy, sample, outcome family, or estimability rule requires a dated deviation memo approved by the PI before execution. First-run results are retained and reported. The `analysis/outcomes/` guard remains fail-closed.

11.2 Deviation log

All entries below are pre-tag amendments to a draft. None was made with access to outcome data: the seal was closed throughout, `dax/data_raw/` held no extract, and every quantity consulted was either date arithmetic on the frozen registry or a power statistic on a synthetic fixture stamped `NOT_EVIDENCE_SYNTHETIC_SMOKE_TEST`. No estimated treatment effect existed or was examined.

Date	Ref	Change	Counter-signed
2026-08-18	D1	Primary specification changed from a stacked event study to a continuous cumulative-dose design; stack demoted to secondary. Decisions 1, 4, 5 re-scoped/reversed/retired; Decision 2 moved.	2026-08-18
2026-08-18	D3	Power pass bar changed from `min(6.5pp, 0.5 x baseline_gap)` to a frozen absolute constant computed once over the pre-event window.	2026-08-18
2026-08-18	D4	Primary estimand named as an incumbent margin; entrant-margin companion registered as a secondary design with a frozen pre-period entry mix.	2026-08-18
2026-08-18	F2	Five event rows demoted to `pending_second_date_locator`; `date_conflict` column added; the release-dating rule is now enforced by `validate_event_registry.py` rather than applied by hand.	2026-08-18
2026-08-18/21	A5	Entrant companion demoted from registered secondary to exploratory after the pre-event audit found `pi_go` non-estimable at the registered level; removed from Gate-1 power and confirmatory claims.	Evidence-triggered red-team adjudication; restoration NEED_HUMAN

Decision numbering is never reused for a different purpose without saying so. Where D1 changed a decision's role, the original number is retained and the change is stated at the point of use, so the 2026-08-06 approval record remains readable against this draft.

References and source register

- Callaway, Goodman-Bacon, and Sant'Anna (2024), continuous-treatment DiD.
- Callaway and Sant'Anna (2021), multiple-period DiD.
- Rambachan and Roth (2023), parallel-trends sensitivity.
- DAX proposal and execution plan in `docs/`.
- Signed feasibility evidence in `dax/memo/feasibility_note.md` and

`dax/memo/w05_legwork_2026-07-10.md`.

- OpenAI API changelog and

deprecations, retrieved 2026-08-06.